

CHAPTER 4

ARE SETS ALL THERE IS?

Our next goal is to determine whether the basic results of naive set theory in Chapter 2 can be proved on the basis of the axioms of Zermelo. Right at the start we hit a snag: to define the crucial notion of *equinumerosity* we need functions; to define *countable sets* we need the specific set $\mathbb{N}$ of natural numbers; the fundamental theorem 2.21 of Cantor is about the set $\mathbb{R}$ of real numbers, etc. Put another way, the results of Chapter 2 are not only about sets, but about points, numbers, functions, Cartesian products and many other mathematical objects which are plainly not sets. Where will we find these objects in the axioms of Zermelo which speak only about sets?

An obvious solution is to assume that these non-sets are among the *atoms* which are allowed by Zermelo's theory and to add axioms which express our basic intuitions about points, numbers, functions, etc. This is possible but awkward and there is a much better solution.

A typical example of the method we will adopt is the “identification” of the (directed) geometric line Π with the set $\mathbb{R}$ of real numbers, via the correspondence which “identifies” each point P on the line with its coordinate $x(P)$ with respect to a fixed choice of an origin O . What is the precise meaning of this “identification”? *Certainly not that points are real numbers*. Men have always had direct geometric intuitions about points which have nothing to do with their coordinates and which existed before Descartes discovered analytic geometry. Every Athenian of the classical period understood the meaning of the sentence

Phaliron is between Piraeus and Sounion along the Saronic coast⁷

even though he was (by necessity) ignorant of analytic geometry. In fact, many educated ancient Athenians had an excellent understanding of the Pythagorean Theorem, without knowing how to coordinatize the plane. What we mean by the “identification” of Π with $\mathbb{R}$ is that the correspondence $P \mapsto x(P)$ gives a **faithful representation** of Π in $\mathbb{R}$ which allows us to give arithmetic definitions for all the useful geometric notions and to study the mathematical properties of Π **as if points were real numbers**. For example, the

⁷These are seaside suburbs of Athens.

quoted sentence above is expressed by the inequalities

$$x(\text{Piraeus}) < x(\text{Phaliron}) < x(\text{Sounion}),$$

assuming that the coordinates increase in the easterly direction. In the same way, we will discover within the universe of sets *faithful representations* of all the mathematical objects we need, and we will study set theory on the basis of the lean axiomatic system of Zermelo **as if all mathematical objects were sets**. The delicate problem in specific cases is to formulate precisely the correct definition of a “faithful representation” and to prove that one such exists.

4.1. Ordered pair. We consider first the basic (**ordered**) **pair** operation. Intuitively, the pair (x, y) of two objects x and y is the “thing” which has a “first member” x and a “second member” y , and it is different from the unordered pair $\{x, y\}$ since (for example) $\{0, 1\} = \{1, 0\}$ while $(0, 1) \neq (1, 0)$. Thus, the first characteristic property of the ordered pair is the following:

$$(OP1) \quad (x, y) = (x', y') \iff x = x' \& y = y'.$$

There is a second, perhaps less obvious characteristic property of pairs which makes it possible to define Cartesian products: for any two sets A, B ,

$$(OP2) \quad \text{the class } A \times B =_{\text{df}} \{(x, y) \mid x \in A \& y \in B\} \text{ is a set.}$$

Thus, the problem of representing the notion of “pair” in set theory takes the following precise form: we must define a definite operation (x, y) such that (OP1) and (OP2) follow from the axioms of Zermelo.

4.2. Lemma. *The Kuratowski pair operation*

$$(x, y) =_{\text{df}} \{\{x\}, \{x, y\}\} \tag{4-1}$$

has properties (OP1) and (OP2).

PROOF. (OP1). The direction $\Leftarrow$ is obvious. For the non-trivial direction $\Rightarrow$, we distinguish two cases.

If $x = y$, then $\{x, y\} = \{x, x\} = \{x\}$, the set $(x, y) = \{\{x\}, \{x\}\} = \{\{x\}\}$ is a singleton, hence the set (x', y') which is assumed equal to it is also a singleton, so that $x' = y'$ and $(x', y') = \{\{x'\}\}$; and since this last singleton is equal to $\{\{x\}\}$, we have $x = x'$ and, hence, also $y = x = x' = y'$.

If $x \neq y$, then the members of (x, y) are the singleton $\{x\}$ and the doubleton $\{x, y\}$, and these must correspond with the members $\{x'\}$ and $\{x', y'\}$ of the equal set (x', y') , so that we must have $\{x\} = \{x'\}$, $\{x, y\} = \{x', y'\}$, and then, immediately, $x = x'$ and $y = y'$.

(OP2). It is enough to prove that for any two sets A, B , there is a set C such that

$$x \in A \& y \in B \implies \{\{x\}, \{x, y\}\} \in C; \tag{4-2}$$

because then

$$A \times B = \{z \in C \mid (\exists x \in A)(\exists y \in B)[z = (x, y)]\},$$

and $A \times B$ is a set by the Subset Axiom (III). To prove (4-2), we compute:

$$\begin{aligned} x \in A, y \in B &\implies \{x\}, \{x, y\} \subseteq (A \cup B) \\ &\implies \{x\}, \{x, y\} \in \mathcal{P}(A \cup B) \\ &\implies \{\{x\}, \{x, y\}\} \subseteq \mathcal{P}(A \cup B) \\ &\implies (x, y) \in \mathcal{P}(\mathcal{P}(A \cup B)), \end{aligned}$$

so that we can take $C = \mathcal{P}(\mathcal{P}(A \cup B))$. $\dashv$

We now fix a specific definite operation (x, y) which satisfies (OP1) and (OP2), perhaps the Kuratowski pair defined in the proof of 4.2, perhaps some other: from now on we may forget the specific definition chosen, the only thing that counts is that the pair operation satisfies (OP1) and (OP2).

4.3. Exercise. *Let*

$$\begin{aligned} \text{Pair}(z) &\iff_{\text{df}} (\exists x)(\exists y)[z = (x, y)], \\ \text{First}(z) &=_{\text{df}} \begin{cases} \text{the unique } x \text{ such that } (\exists y)[z = (x, y)], & \text{if } \text{Pair}(z), \\ z, & \text{otherwise,} \end{cases} \\ \text{Second}(z) &=_{\text{df}} \begin{cases} \text{the unique } y \text{ such that } (\exists x)[z = (x, y)], & \text{if } \text{Pair}(z), \\ z, & \text{otherwise.} \end{cases} \end{aligned}$$

It follows that

$$\text{Pair}(z) \iff z = (\text{First}(z), \text{Second}(z)).$$

Using the ordered pair we can easily define triples, quadruples, etc., as well as the corresponding products, e.g.,

$$(x, y, z) =_{\text{df}} (x, (y, z)), \quad (4-3)$$

$$(x, y, z, w) =_{\text{df}} (x, (y, (z, w))) = (x, (y, (z, w))), \quad (4-4)$$

$$A \times B \times C =_{\text{df}} A \times (B \times C), \quad (4-5)$$

etc. By this definition, a tuple of length $n + 1$ is a pair with second member a tuple of length n .

4.4. Exercise. *For all* x, y, z, x', y', z' ,

$$(x, y, z) = (x', y', z') \iff x = x' \ \& \ y = y' \ \& \ z = z'.$$

4.5. Disjoint union. For each set A and a fixed object blue, we can think of the set of pairs $\{\text{blue}\} \times A$ as a “blue copy” of A , the act of replacing each $a \in A$ by the pair (blue, a) being the set theoretic equivalent of painting a blue. We fix two such distinct “colors”,

$$\text{blue} =_{\text{df}} \emptyset, \quad \text{white} =_{\text{df}} \{\emptyset\}, \quad (4-6)$$

and we define the **disjoint union** of two sets by

$$A \uplus B =_{\text{df}} (\{\text{blue}\} \times A) \cup (\{\text{white}\} \times B).$$

The construction is pictured in Figure 4.1, and the notion is useful, as we will see. It should be clear that the specific identity of blue and white must be deliberately and instantly forgotten: all that matters is that blue $\neq$ white.

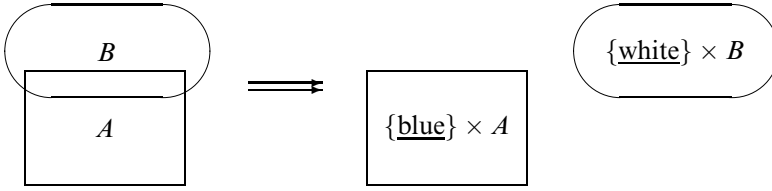


FIGURE 4.1. Constructing the disjoint union.

4.6. Exercise. Show that for all sets A, B , $A \uplus \emptyset \subseteq A \uplus B$. Assuming that we use the Kuratowski pair to define products, give an example where the plausible inclusion $A \subseteq A \uplus B$ is not true.

Next we consider the notion of **relation** which permeates mathematics. Intuitively, a *binary relation* R between objects $x \in A$ and $y \in B$ is a condition which is satisfied by some $x \in A, y \in B$ and fails for others. For example, the relation

$$x R y \iff_{\text{df}} x \text{ is a son of } y$$

is defined on $A = \{\text{men}\}$, $B = \{\text{women}\}$ and holds for x, y precisely if y has given birth to x . The obvious way to represent a binary relation in set theory is to identify it with its *extension*, the set of pairs which satisfy it.

4.7. Relations. A **binary relation** on the sets A, B is any subset R of the Cartesian product $A \times B$. We will use synonymously the notations

$$R(x, y) \iff x R y \iff (x, y) \in R.$$

Obvious examples of binary relations are the **identity** and the relations of **membership** and **subsethood** restricted to some set A ,

$$x =_A y \iff_{\text{df}} x \in A \ \& \ y \in A \ \& \ x = y,$$

$$x \in_A Y \iff_{\text{df}} x \in A \ \& \ Y \subseteq A \ \& \ x \in Y,$$

$$X \subseteq_A Y \iff_{\text{df}} X \subseteq Y \subseteq A,$$

which by the definition are represented respectively by the sets

$$=_A =_{\text{df}} \{(x, y) \in A \times A \mid x = y\},$$

$$\in_A =_{\text{df}} \{(x, Y) \in A \times \mathcal{P}(A) \mid x \in Y\},$$

$$\subseteq_A =_{\text{df}} \{(X, Y) \in \mathcal{P}(A) \times \mathcal{P}(A) \mid X \subseteq Y\}.$$

4.8. Relations and definite conditions. The definite conditions of identity $=$, membership $\in$ and subsethood $\subseteq$ on the domain $\mathcal{W}$ of all objects are not relations according to 4.7, in fact they are not even “coextensive” with sets of pairs, because their extensions are “too large”. This is important, the distinction between relations and definite conditions: briefly, *every relation determines a definite condition but (in general) the converse does not hold*. The precise situation is detailed in the next Exercise. In practice we will often refer to *the relation $=$ on the set A* , meaning (without possibility of confusion) the restriction $=_A$ as we just defined it.

4.9. Exercise. (1) For every binary relation $R \subseteq (A \times B)$, the condition

$$R^*(x, y) \iff_{\text{df}} (x, y) \in R$$

is definite. (Nothing to prove here, unless you want to practice applying 3.18.)

(2) For every binary definite condition P and any two sets A, B , the **restriction**

$$P_{A,B} =_{\text{df}} \{(x, y) \in A \times B \mid P(x, y)\}$$

of P to $A \times B$ is a binary relation on A, B .

4.10. Properties of relations on a set A . Binary relations with both arguments ranging over the same set are especially important, and they are classified and studied according to the structural properties they may enjoy. Here are three such properties which come up often, in various combinations, where $P \subseteq A \times A$:

$$P \text{ is reflexive} \iff_{\text{df}} (\forall x \in A)[x P x],$$

$$P \text{ is symmetric} \iff_{\text{df}} (\forall x, y \in A)[x P y \implies y P x],$$

$$P \text{ is transitive} \iff_{\text{df}} (\forall x, y, z \in A)[[x P y \& y P z] \implies x P z].$$

We call P an **equivalence relation** on A if it has all three of these properties.

Equivalence relations are very useful and we will meet examples of them in practically every chapter of these Notes. They are often denoted by symbols like $\sim, \approx, \simeq$, so that their three characteristic properties take the form

$$x \sim x,$$

$$x \sim y \implies y \sim x,$$

$$[x \sim y \& y \sim z] \implies x \sim z.$$

4.11. Exercise. On each set A , the identity relation $\{(x, y) \mid x = y \in A\}$, the identically true relation $\{(x, y) \mid x, y \in A\}$ and for each $B \subseteq A$ the relation

$$x \sim_{A/B} y \iff_{\text{df}} x = y \in A \vee [x \in B \& y \in B]$$

are all equivalence relations.

4.12. Proposition. Suppose $\sim$ is an equivalence relation on the set A , let for each $x \in A$,

$$[x/\sim] = \{y \in A \mid x \sim y\} \tag{4-7}$$

be the **equivalence class**⁸ of x , and let

$$\llbracket A/\sim \rrbracket = \{[x/\sim] \in \mathcal{P}(A) \mid x \in A\} \tag{4-8}$$

be the set of all equivalence classes. It follows that $[x/\sim] \neq \emptyset$ for each $x \in A$, and for all $x, y \in A$,

$$x \sim y \iff [x/\sim] = [y/\sim], \tag{4-9}$$

$$x \not\sim y \iff [x/\sim] \cap [y/\sim] = \emptyset, \tag{4-10}$$

⁸Each $[x/\sim]$ is obviously a set, a subset of A , and it would be more appropriately named the “equivalence set” of x , but the classical terminology goes way back and has been frozen.

so that $\llbracket A/\sim \rrbracket$ is a family of non-empty and pairwise disjoint subsets of A such that $\bigcup \llbracket A/\sim \rrbracket = A$.

Conversely, for each family $\mathcal{E}$ of non-empty and pairwise disjoint subsets of A such that $A = \bigcup \mathcal{E}$, the relation

$$x \sim y \iff_{\text{df}} (\exists X \in \mathcal{E})[x \in X \ \& \ y \in X]$$

is an equivalence relation on A and $\llbracket A/\sim \rrbracket = \mathcal{E}$.

PROOF. Each $[x/\sim] \neq \emptyset$, since $x \in [x/\sim]$. By the transitivity and symmetry of $\sim$,

$$t \sim x \ \& \ x \sim y \implies t \sim y, \quad t \sim y \ \& \ x \sim y \implies t \sim x$$

so that

$$\begin{aligned} x \sim y &\implies (\forall t \in A)[t \sim x \iff t \sim y] \\ &\implies [x/\sim] = [y/\sim]. \end{aligned}$$

This implies immediately both (4-9) and (4-10). For the converse, the reflexivity and symmetry of $\sim$ are trivial. If $x \sim y$ and $y \sim z$, then there exist sets X, Y in $\mathcal{E}$ such that $x, y \in X$, $y, z \in Y$, so in particular $y \in X \cap Y$ and since the sets in $\mathcal{E}$ are pairwise disjoint, we have $X = Y$, so $x \sim z$. $\dashv$

4.13. Exercise. What are the equivalence classes of the equivalence relations in Exercise 4.11?

Following up the same idea, we identify each **ternary relation** R on the sets A, B, C with the set of triples which satisfy it, so that a ternary relation on A, B, C is simply an arbitrary subset of $A \times B \times C$. We will use synonymously the notations

$$R(x, y, z) \iff_{\text{df}} (x, y, z) \in R.$$

As with relations, we represent functions in set theory by identifying them with their “graphs”.

4.14. Functions. A **function** (or **mapping** or **transformation**) $f : A \rightarrow B$ with domain the set A and range the set B is any subset $f \subseteq (A \times B)$ which satisfies the condition

$$(\forall x \in A)(\exists! y \in B)[(x, y) \in f],$$

in more detail,

$$(\forall x \in A)(\exists y \in B)[(x, y) \in f],$$

$$\text{and } (x, y) \in f \ \& \ (x, y') \in f \implies y = y'.$$

If we picture the product $A \times B$ in the plane as in Figure 4.2, this means that every “vertical line” intersects the set f in exactly one point: for $x \in A$ and $f : A \rightarrow B$, we will write, as usual,

$$\begin{aligned} f(x) &=_{\text{df}} \text{the unique } y \in B \text{ such that } (x, y) \in f & (x \in A). \\ &= \text{the value of } f \text{ on the element } x \end{aligned}$$

The function space

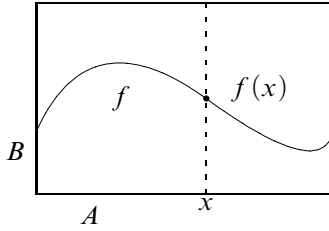


FIGURE 4.2. A function as a set of ordered pairs.

$$\begin{aligned} (A \rightarrow B) &=_{\text{df}} \{f \subseteq A \times B \mid f : A \rightarrow B\} \\ &= \{f \in \mathcal{P}(A \times B) \mid (\forall x \in A)(\exists! y \in B)(x, y) \in f\} \end{aligned}$$

is a set by the Subset Axiom (III).

We will use all the familiar notations and abbreviations in connection with functions, e.g., sometimes writing the argument without the parentheses or as an index,

$$f(x) = fx = f_x.$$

The $\mapsto$ notation is also useful; for example, an **indexed family of sets** is a function

$$A = (i \mapsto A_i)_{i \in I} : I \rightarrow E$$

for some $I \neq \emptyset$ and some E , where each A_i is a set. We refer to I as the **index set** and we define the union and intersection of the family in the usual way,

$$\begin{aligned} \bigcup_{i \in I} A_i &=_{\text{df}} \{x \in \bigcup E \mid (\exists i \in I)[x \in A_i]\}, \\ \bigcap_{i \in I} A_i &=_{\text{df}} \{x \in \bigcup E \mid (\forall i \in I)[x \in A_i]\}. \end{aligned} \quad (4-11)$$

We can also define the **product** of an indexed family, the set of functions which select for each $i \in I$ one element from the value A_i ,

$$\prod_{i \in I} A_i =_{\text{df}} \{f : I \rightarrow \bigcup_{i \in I} A_i \mid (\forall i \in I)[f(i) \in A_i]\}. \quad (4-12)$$

Injections, surjections, bijections (correspondences), images and pre-images of functions are defined exactly as in the Introduction. We will be using the following notations for the relevant function spaces:

$$\begin{aligned} (A \hookrightarrow B) &=_{\text{df}} \{f \in (A \rightarrow B) \mid f \text{ is an injection, one-to-one}\}, \\ (A \twoheadrightarrow B) &=_{\text{df}} \{f \in (A \rightarrow B) \mid f \text{ is a surjection, onto } B\}, \\ (A \xrightarrow{\sim} B) &=_{\text{df}} (A \hookrightarrow B) \cap (A \twoheadrightarrow B) \text{ (bijection, correspondence)}. \end{aligned}$$

For each $X \subseteq A$, the **restriction** $f \upharpoonright X$ of a function $f : A \rightarrow B$ is obtained by cutting f down so it is defined only on X ,

$$f \upharpoonright X =_{\text{df}} \{(x, y) \in f \mid x \in X\}. \quad (4-13)$$

It is also useful to notice that the basic condition of “functionhood”

$$\text{Function}(f) \iff_{\text{df}} (\exists A)(\exists B)[f \in (A \rightarrow B)] \quad (4-14)$$

is evidently definite. When we refer to a *function* f without identifying specific sets A, B such that $f : A \rightarrow B$, we will mean any set f which satisfies the condition $\text{Function}(f)$.

4.15. Exercise. *Prove from the axioms, that for each function f , the domain of definition of f*

$$\text{Domain}(f) =_{\text{df}} \{x \mid (\exists y)[(x, y) \in f]\}$$

and the image of f

$$\text{Image}(f) =_{\text{df}} \{y \mid (\exists x)[(x, y) \in f]\}$$

are sets, and for each set B ,

$$\text{if } \text{Image}(f) \subseteq B, \text{ then } f : \text{Domain}(f) \rightarrow B.$$

As a consequence,

$$\text{if } \text{Function}(f), \text{ then } f : \text{Domain}(f) \rightarrow \text{Image}(f).$$

4.16. About functions as sets of ordered pairs. This “identification” of a function $f : A \rightarrow B$ with the set of pairs $\{(x, y) \in A \times B \mid f(x) = y\}$ has generated some controversy, because we have natural “operational” intuitions about functions and by “function” we often mean a formula or a rule of computation. For example, the two functions on the reals

$$\begin{aligned} f(x, y) &=_{\text{df}} (x + y)^2, \\ g(x, y) &=_{\text{df}} x^2 + 2xy + y^2 \end{aligned}$$

are identified in set theory, although they are obviously different as computation rules. There is no problem with this if we keep clear in our minds that the “definition” 4.14 does not replace the intuitive notion of function but only represents it within *set theory*, faithfully for the uses to which we put this notion *within set theory*:⁹ first (and foremost), to define equinumerosity and the size comparison condition for sets with no reference to objects outside axiomatic set theory,

$$A =_c B \iff_{\text{df}} (\exists f)[f : A \twoheadrightarrow B] \iff (A \twoheadrightarrow B) \neq \emptyset,$$

$$A \leq_c B \iff_{\text{df}} (\exists f)[f : A \rightarrow B] \iff (A \rightarrow B) \neq \emptyset.$$

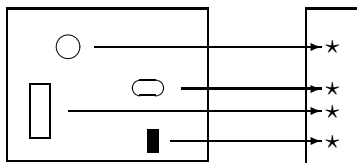
$$A <_c B \iff_{\text{df}} A \leq_c B \ \& \ A \neq_c B.$$

4.17. Exercise. *Prove from the axioms that $A =_c B \implies \mathcal{P}(A) =_c \mathcal{P}(B)$, and identify which of the axioms you are using.*

4.18. Exercise. *Prove from the axioms that if $A =_c A'$ and $B =_c B'$, then*

$$A \uplus B =_c A' \uplus B', \quad A \times B =_c A' \times B', \quad (A \rightarrow B) =_c (A' \rightarrow B').$$

⁹Whether the intuitive notion of function-as-computation-rule can also be represented faithfully in set theory is an interesting problem, for which there does not exist yet a generally accepted solution.

FIGURE 4.3. Cantor's construction of $|A|$ for a four-element set.

4.19. Cantor's notion of cardinal numbers. Ironically, one of the most difficult, intuitive mathematical notions to represent faithfully in set theory is that of *cardinal number*, a most basic concept of the subject. Here is how Cantor introduced it in the same 1895 paper from which we quoted the “definition” of sets in the Introduction:

Every set A has a definite ‘power’, which we will also call its ‘cardinal number’.

We will call by the name ‘power’ or ‘cardinal number’ of A the general concept, which by means of our active faculty of thought, arises from the set A when we make abstraction of its various elements x and of the order in which they are given.

We denote the result of this double act of abstraction, the cardinal number or power of A by $\overline{\overline{A}}$. Since every element x , if we abstract from its nature becomes a ‘unit’, the cardinal number $\overline{\overline{A}}$ is a definite set composed of units, and this number has existence in our minds as an intellectual image or projection of the given set A .

After some discussion, Cantor infers from this “definition” that cardinal numbers have the following two fundamental properties:

$$A =_c \overline{\overline{A}},$$

$$\text{if } A =_c B, \text{ then } \overline{\overline{A}} = \overline{\overline{B}}.$$

The first of these flows quite naturally from Cantor's conception: the process of abstraction which associates with each $x \in A$ a corresponding “unit” u_x evidently defines a correspondence $x \mapsto u_x$ between A and $\overline{\overline{A}}$. Cantor gives a brief argument for the second whose key phrase is that

$\overline{\overline{A}}$ grows, so to speak, out of A in such a way that from every element x of A a special unit of A arises.

To get the second property out of this we must assume that the “special units of A ” depend only on “how many” members A has and not the nature of these members, which begs the question of cardinality, but there it is.

There is a third, more technical property of cardinal numbers, which Cantor uses routinely with no special mention to define and study operations which act on infinite families of sets: *for every family of sets $\mathcal{E}$, $\{\overline{\overline{X}} \mid X \in \mathcal{E}\}$ is a set.* Thus, however we understand Cantor's construction, it is quite clear what we

must do to represent it faithfully in set theory. We substitute modern notation for Cantor's awkward double bar symbolism.

4.20. Problem of Cardinal Assignment: *to define an operation $|A|$ on the class of sets which satisfies*

$$(C1) \quad A =_c |A|,$$

$$(C2) \quad \text{if } A =_c B, \text{ then } |A| = |B|,$$

$$(C3) \quad \text{for each set of sets } \mathcal{E}, \{ |X| \mid X \in \mathcal{E} \} \text{ is a set.}$$

The problem is quite difficult and it was not solved until the twenties, by von Neumann, whose elegant construction uses both the Axioms of Choice and Replacement. We will present it in Chapter 12, as the culmination of a lot of work. In the meantime, notice that there are plenty of definite operations which satisfy (C1) and (C3), including the obvious $|A| = A$! And as it turns out, these two properties suffice for the development of a very satisfactory theory of cardinality.

4.21. Cardinal numbers. A (weak) **cardinal assignment** is any definite operation on sets $A \mapsto |A|$ which satisfies (C1), and (C3), and it is a **strong cardinal assignment** if it also satisfies (C2). The **cardinal numbers** (relative to a given cardinal assignment) are its values,

$$(C4) \quad \text{Card}(\kappa) \iff \kappa \in \text{Card} \iff_{\text{df}} (\exists A)[\kappa = |A|].$$

4.22. Exercise. *Prove that for any cardinal assignment and any two sets A, B ,*

$$|A| = |B| \implies A =_c B,$$

so that the converse of (C2) is true of all cardinal assignments, including the weak ones.

We fix one, specific (possibly weak) cardinal assignment and we define the arithmetical operations on the cardinals as follows:

$$\begin{aligned} \kappa + \lambda &=_{\text{df}} |\kappa \uplus \lambda| &=_{\text{c}} \kappa \uplus \lambda, \\ \kappa \cdot \lambda &=_{\text{df}} |\kappa \times \lambda| &=_{\text{c}} \kappa \times \lambda, \\ \kappa^\lambda &=_{\text{df}} |(\lambda \rightarrow \kappa)| &=_{\text{c}} (\lambda \rightarrow \kappa). \end{aligned}$$

The infinitary operations are defined similarly:¹⁰

$$\begin{aligned} \sum_{i \in I} \kappa_i &=_{\text{df}} |\{(i, x) \in I \times \bigcup_{i \in I} \kappa_i \mid x \in \kappa_i\}|, \\ \prod_{i \in I} \kappa_i &=_{\text{df}} |\prod_{i \in I} \kappa_i|. \end{aligned}$$

The motivation is clear, e.g., the sum $\kappa + \lambda$ is the “number of elements” in the set we get by putting together disjoint copies of κ and λ .

¹⁰It is traditional to use the cap Greek Π to denote both the Cartesian product of sets and the cardinal operation of infinite product, and it does not really cause any confusion.

Notice that there is only one choice for $|\emptyset|$,

$$0 =_{\text{df}} |\emptyset| = \emptyset, \quad (4-15)$$

since only $|\emptyset| = \emptyset$ satisfies $\emptyset =_c |\emptyset|$. It is also convenient to set

$$1 =_{\text{df}} |\{0\}|, \quad 2 =_{\text{df}} |\{0, 1\}| \quad (4-16)$$

so we have handy names for the cardinal numbers of singletons and doubletons.

4.23. Exercise. For all sets A, B , $|A \cup B| \leq_c |A| + |B|$, and

$$\text{if } A \cap B = \emptyset, \text{ then } |A \cup B| =_c |A| + |B|.$$

4.24. Exercise. For all cardinal numbers, if $\kappa_1 =_c \kappa_2$ and $\lambda_1 =_c \lambda_2$, then

$$\kappa_1 + \lambda_1 =_c \kappa_2 + \lambda_2, \quad \kappa_1 \cdot \lambda_1 =_c \kappa_2 \cdot \lambda_2, \quad \kappa_1^{\lambda_1} =_c \kappa_2^{\lambda_2}.$$

4.25. Cardinal arithmetic. It looks quite silly to develop the theory of a weak cardinal assignment which could be just the identity $|X| = X$, but the notation of cardinal numbers and the arithmetical operations on them is useful for expressing simply complex “equinumerosities”. Consider the formula

$$\kappa^{(\lambda+\mu)} =_c \kappa^\lambda \cdot \kappa^\mu. \quad (4-17)$$

It looks obvious, it is true by Exercise 4.28, and it expresses exactly the same fact as

$$((\lambda \uplus \mu) \rightarrow \kappa) =_c (\lambda \rightarrow \kappa) \times (\mu \rightarrow \kappa), \quad (4-18)$$

more simply, or so some would say. More significantly:

(1) the systematic development of formulas like (4-17) leads to a *cardinal arithmetic* which in the end suggests new (and useful) facts about equinumerosities by analogy with ordinary arithmetic; and

(2) when we do construct von Neumann’s strong cardinal assignment, we will have already proved all the interesting facts about cardinals with $=_c$ in place of $=$: all we will need to do is remove in our minds the subscript $_c$ from facts we already understand, because of the following, simple fact.

4.26. Exercise. If cardinal numbers are defined using a strong cardinal assignment, then for all cardinals κ, λ ,

$$|\kappa| = \kappa \text{ and } \kappa =_c \lambda \iff \kappa = \lambda. \quad (4-19)$$

The basic technique for proving identities of cardinal arithmetic is to use systematically (C1) and the replacement properties of Exercise 4.18. To prove the associativity of cardinal addition, for example, we compute:

$$\begin{aligned} \kappa + (\lambda + \mu) &= \kappa \uplus (\lambda + \mu) && \text{by def.,} \\ &= \kappa \uplus (\lambda \uplus \mu) && \text{by def., (C1) and 4.18,} \\ &= (\kappa \uplus \lambda) \uplus \mu && \text{by a direct argument,} \\ &= (\kappa + \lambda) + \mu && \text{reversing the steps.} \end{aligned}$$

The mathematical essence of the proof is the alleged “direct argument”, which in this case is quite easy:

4.27. Exercise. Prove that for any three sets K, L, M ,

$$K \uplus (L \uplus M) =_c (K \uplus L) \uplus M.$$

4.28. Exercise. Prove (4-17), by showing first that for any three sets K, L, M ,

$$\text{if } L \cap M = \emptyset, \text{ then } ((L \cup M) \rightarrow K) =_c (L \rightarrow K) \times (M \rightarrow K).$$

To see how the more technical condition (C3) comes into play, consider the equation

$$|\bigcup_{i \in I} A_i| =_c \sum_{i \in I} |A_i|, \quad (4-20)$$

which should certainly be true when the sets in the family $(i \mapsto A_i)_{i \in I}$ are pairwise disjoint. To make sense of it, before trying to prove it, we must know that *there is a function* $(i \mapsto |A_i|)$, and the proof of this requires (C3), as follows:

4.29. Lemma. For every indexed family of sets $A = (i \mapsto A_i)_{i \in I}$, there exists a function $f : I \rightarrow f[I]$ such that

$$f(i) = |A_i| \quad (i \in I).$$

PROOF. By (C3) with

$$\mathcal{E} = \{A_i \mid i \in I\} = A[I],$$

there exists a set W which contains every $|A_i|$ for $i \in I$, and we can set

$$f =_{\text{df}} \{(i, w) \in I \times W \mid w = |A_i|\}. \quad \dashv$$

As it happens, equations like (4-20) cannot be proved without the Axiom of Choice, so we will have little need of (C3) before Chapter 8.

4.30. Structured sets. A *topological space* is a set X of *points* endowed with a *topological structure*, which is determined by a collection $\mathcal{T}$ of subsets of X satisfying the following three properties:

1. $\emptyset, X \in \mathcal{T}$.
2. $A, B \in \mathcal{T} \implies A \cap B \in \mathcal{T}$.
3. For every family $\mathcal{E} \subseteq \mathcal{T}$ of sets in $\mathcal{T}$, the unionset $\bigcup \mathcal{E}$ is also in $\mathcal{T}$.

A family of sets $\mathcal{T}$ with these properties is called a **topology** on X , with **open sets** its members and **closed sets** the complements of open sets relative to X , i.e., all $X \setminus G$ with G open.

Notions like this of sets “endowed” with structure abound in mathematics: there are graphs, groups, vector spaces, sheaves, manifolds, partially ordered sets, etc. etc. In each of these cases we have a set X , typically called “the space”, and a complex of related objects which impose a structure on the space—functions, families of sets, other spaces with their own structure, etc. The pairing operation provides a simple and flexible way to model such notions faithfully in set theory.

A **structured set** is a pair

$$U = (A, \mathcal{S}), \quad (4-21)$$

where $A = \text{Field}(U)$ is a set, the *field* or *space* of U , and $\mathcal{S}$ is an arbitrary object, the *frame*¹¹ of U .

For example, a **topological space** is a structured set $(X, \mathcal{T})$, where the frame $\mathcal{T}$ is a topology on X , as above. A *group* is a structured set

$$U = (G, (e, \cdot)) \quad (4-22)$$

where $e \in G$ and $\cdot : G \times G \rightarrow G$ is a binary function, satisfying the *group axioms*, which do not concern us here. Notice that by the definition of triple (4-3), definition (4-22) is equivalent to

$$U = (G, e, \cdot). \quad (4-23)$$

It is quite common that the frame of a structured set is a tuple of objects, and then the structured set is also a tuple, with its field as the first element. We will meet numerous examples of this in the sequel.

Following usual mathematical practice, we will systematically confuse a structured set with its field when the frame is understood from the context or is not relevant. For example, we will refer to “the topological space X ” rather than “ $(X, \mathcal{T})$ ”, with “points” the members of X , “subsets” the subsets of X , etc. In the general case, the members of a structured set U are the members of $\text{Field}(U)$,

$$x \in U \iff_{\text{df}} x \in \text{Field}(U), \quad (4-24)$$

the subsets of U are the subsets of $\text{Field}(U)$, etc. Notice that the terminological convention (4-24) cannot possibly cause a misunderstanding: since we have (deliberately) not settled on a specific pairing operation—and have even left open the possibility that $(A, \mathcal{S})$ may be an atom (!)—the statement

$$x \in (A, \mathcal{S})$$

cannot possibly mean anything until we define it, and we just did this by (4-24).

Problems for Chapter 4

The definition of ordered pair in the proof of 4.2 is due to the Polish set theorist and topologist Kuratowski. A few years before Kuratowski’s construction, the American analyst Wiener had discovered the following, somewhat more complex but interesting solution of this problem.

x4.1. (Wiener) The properties (OP1) and (OP2) in 4.1 hold with the following definition of pair:

$$(x, y) =_{\text{df}} \{\{\emptyset, \{x\}\}, \{\{y\}\}\}.$$

¹¹It would be more suggestive to call $\mathcal{S}$ *the structure* of the structured set $(A, \mathcal{S})$, but the word is heavily overloaded in logic and set theory and it is best to avoid attaching it to one more precise notion. Some people call “structures” what we have called “structured sets” here, at least when they are simple enough.

x4.2. The properties (OP1) and (OP2) in 4.1 hold with the following definition of pair:

$$(x, y) =_{\text{df}} \{\{0, x\}, \{1, y\}\},$$

where 0, 1 are two distinct objects, e.g., those defined by (4-16). **HINT.** Compute $(0, 0)$, $(0, 1)$, $(1, 1)$ and $(1, 0)$ (with this pair) to see what goes on, and then take cases on whether $x = y$ or not.

x4.3. Prove from the axioms that for all sets A, B, C ,

$$((A \times B) \rightarrow C) =_c (A \rightarrow (B \rightarrow C)).$$

x4.4. Prove from the axioms the theorem of Cantor 2.21, that for every set A , $A <_c \mathcal{P}(A)$. Which axioms do you need?

x4.5. A binary relation $\sim \subseteq (A \times A)$ is an equivalence relation on A if and only if there exists some set Q and a surjection

$$\pi : A \twoheadrightarrow Q \tag{4-25}$$

such that

$$x \sim y \iff \pi(x) = \pi(y). \tag{4-26}$$

When (4-25) and (4-26) hold, we call Q a **quotient** of A by $\sim$ and π a **determining surjection** of $\sim$. The proof of 4.12 yields the quotient $\llbracket A/\sim \rrbracket$ and the determining surjection $(x \mapsto [x/\sim])$, but in specific cases there exist other quotients which help us understand better the structure of the equivalence relation at hand.

x4.6. Suppose $\sim$ is an equivalence relation on A and $\pi : A \rightarrow A$ satisfies

$$x \sim y \implies \pi(x) = \pi(y) \in [x/\sim].$$

Prove that π is a determining surjection witnessing that its image $\pi[A] \subseteq A$ is a quotient of A by $\sim$.

x4.7. Fix an element $x_0 \in A$ in some set and define on the function space $(A \rightarrow B)$ the relation

$$f \sim g \iff_{\text{df}} f(x_0) = g(x_0).$$

Prove that $\sim$ is an equivalence relation and find a determining surjection $\pi : (A \rightarrow B) \twoheadrightarrow B$ which witnesses that B is a quotient of $(A \rightarrow B)$ by $\sim$.

x4.8. Let $x_0 \neq x_1$ be two distinct elements in A and find a determining surjection which witnesses that $(B \times B)$ is a quotient of $(A \rightarrow B)$ by the equivalence relation

$$f \sim g \iff_{\text{df}} f(x_0) = g(x_0) \ \& \ f(x_1) = g(x_1).$$

x4.9. Suppose $\sim$ is an equivalence relation on A and $f : A \rightarrow A$ is a function which **respects** $\sim$, i.e.,

$$x \sim y \implies f(x) \sim f(y).$$

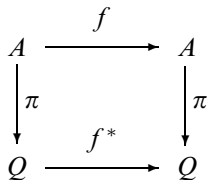


FIGURE 4.4.

Let Q be any quotient of A by $\sim$. Prove that there exists a unique function $f^* : Q \rightarrow Q$ such that the diagram in Figure 4.4 *commutes*, i.e., $\pi f = f^* \pi$,

$$f^*(\pi x) = \pi(f(x)), \quad (x \in A),$$

where $\pi : A \twoheadrightarrow Q$ is a determining surjection.

x4.10. For all cardinal numbers, κ, λ, μ ,

$$\kappa + 0 =_c \kappa, \quad \kappa \cdot 0 =_c 0, \quad \kappa \cdot 1 =_c \kappa.$$

x4.11. For all cardinal numbers κ, λ, μ ,

$$\begin{aligned}
 \kappa + (\lambda + \mu) &= _c (\kappa + \lambda) + \mu, \\
 \kappa + \lambda &= _c \lambda + \kappa.
 \end{aligned}$$

x4.12. For all cardinal numbers κ, λ, μ ,

$$\begin{aligned}
 \kappa \cdot (\lambda \cdot \mu) &= _c (\kappa \cdot \lambda) \cdot \mu, \\
 \kappa \cdot \lambda &= _c \lambda \cdot \kappa, \\
 \kappa \cdot (\lambda + \mu) &= _c \kappa \cdot \lambda + \kappa \cdot \mu.
 \end{aligned}$$

x4.13. For all cardinals κ , $|\mathcal{P}(\kappa)| =_c 2^\kappa$.

x4.14. For all cardinal numbers κ, λ, μ ,

$$\kappa^0 =_c 1, \quad \kappa^1 =_c \kappa, \quad \kappa^2 =_c \kappa \cdot \kappa.$$

x4.15. For all cardinal numbers κ, λ, μ ,

$$\begin{aligned}
 (\kappa \cdot \lambda)^\mu &= _c \kappa^\mu \cdot \lambda^\mu, \\
 \kappa^{(\lambda+\mu)} &= _c \kappa^\lambda \cdot \kappa^\mu, \\
 (\kappa^\lambda)^\mu &= _c \kappa^{\lambda \cdot \mu}.
 \end{aligned}$$

x4.16. For all cardinal numbers κ, λ, μ

$$\begin{aligned}
 \kappa \leq_c \mu &\implies \kappa + \lambda \leq_c \mu + \lambda, \\
 \kappa \leq_c \mu &\implies \kappa \cdot \lambda \leq_c \mu \cdot \lambda, \\
 \lambda \leq_c \mu &\implies \kappa^\lambda \leq_c \kappa^\mu \quad (\kappa \neq 0), \\
 \kappa \leq_c \lambda &\implies \kappa^\mu \leq_c \lambda^\mu.
 \end{aligned}$$

For what values of λ, μ does the third implication fail when $\kappa = 0$?

Caution! As we will see later, strict inequalities between infinite cardinal numbers are not always respected by the algebraic operations, for example, we may have

$$\kappa <_c \mu, \text{ but } \kappa + \lambda =_c \mu + \lambda.$$

x4.17. For all A, B and all cardinals κ, λ ,

$$\prod_{i \in A} B = (A \rightarrow B), \quad \prod_{i \in \lambda} \kappa = \kappa^\lambda.$$

x4.18. Suppose $a \neq b$ are two distinct objects and κ_a, κ_b are cardinals, and prove that

$$\begin{aligned} \kappa_a + \kappa_b &=_c \sum_{i \in \{a,b\}} \kappa_i, \\ \kappa_a \cdot \kappa_b &=_c \prod_{i \in \{a,b\}} \kappa_i. \end{aligned}$$

x4.19. Prove that for all indexed families of cardinals,

$$\kappa \cdot \sum_{i \in I} \lambda_i =_c \sum_{i \in I} \kappa \cdot \lambda_i.$$

x4.20. Show that $\kappa \cdot \lambda = 0 \iff \kappa = 0 \vee \lambda = 0$. Show also one of the directions of the equivalence

$$\prod_{i \in I} \kappa_i = 0 \iff (\exists i \in I)[\kappa_i = 0]. \quad (4-27)$$

(If you think you can show both directions of (4-27), think again and find your error, since one direction requires the Axiom of Choice.)

x4.21. The notion of equivalence according to Zermelo 3.28 coincides with equinumerosity, i.e.,

$$A =_c B \iff A \sim_Z B.$$

The definition 2.6 of infinite and finite sets refers to the set $\mathbb{N}$ of natural numbers and we cannot study these concepts axiomatically before we give a definition of $\mathbb{N}$ directly from the axioms. There is, however, another, simpler definition of these notions which we can give now and which we will later prove (with the Axiom of Choice) equivalent to 2.6.

4.31. Definition. A set A is **infinite according to Dedekind** if there exists an injection

$$f : A \rightarrow B \subsetneq A$$

from A into a proper subset of itself. If A is not Dedekind-infinite, then it is **Dedekind-finite**.

x4.22. If A is Dedekind-infinite and $A =_c B$, then B is also Dedekind-infinite.

x4.23. If A is Dedekind-finite, then every subset of A is also Dedekind-finite.

x4.24. Every set I which satisfies the conditions

$$\emptyset \in I, \quad (\forall x)[x \in I \implies \{x\} \in I]$$

is Dedekind-infinite.

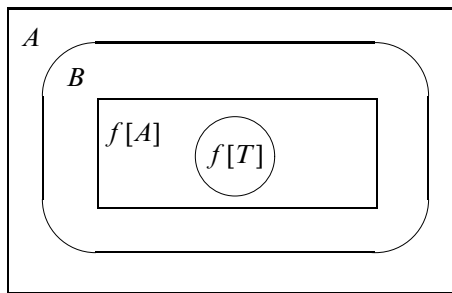


FIGURE 4.5. Zermelo's proof of the Schröder-Bernstein Theorem.

Most of the properties of Dedekind-finite sets require the Axiom of Choice for their proof. Here is one which does not, but it requires some thinking.

***x4.25.** If A is Dedekind-finite and $t \notin A$, then $A \cup \{t\}$ is also Dedekind-finite. **HINT.** Assume $\pi : A \cup \{t\} \rightarrow A \cup \{t\}$ misses some point in $A \cup \{t\}$, and consider cases on whether it misses t ; or $\pi(t) = t$; or “otherwise”, which is where the thinking is required.

The classical proof of the Schröder-Bernstein Theorem 2.26 uses induction on the natural numbers and we cannot justify it now. In the next two problems we outline a very different proof (due to Zermelo), somewhat opaque in its motivation but elegant, short and in no way dependent on the natural numbers.

***x4.26.** If $A' \subseteq B \subseteq A$ and $A =_c A'$, then also $A =_c B$. **HINT.** Suppose $f : A \rightarrow A'$ is a correspondence which witnesses that $A =_c A'$, and

$$Q = B \setminus f[A]$$

is the set of objects in B which are not in the image of A by f . We define the family of subsets of A

$$\mathcal{T} = \{X \mid Q \cup f[X] \subseteq X\}$$

and we first verify that its intersection is a member of it,

$$T =_{\text{df}} \bigcap \mathcal{T} \in \mathcal{T},$$

so that $Q \cup f[T] \subseteq T$. With a bit more work we can show that, in fact, $T = Q \cup f[T]$; this identity then implies that

$$B = T \cup (f[A] \setminus f[T]),$$

which completes the proof, since T and $(f[A] \setminus f[T])$ are disjoint sets and their union is (easily now) equinumerous with A .

***x4.27.** Use Problem x4.26 to give a proof of the Schröder-Bernstein Theorem from the axioms. **HINT.** If $f : A \rightarrow C$ and $g : C \rightarrow A$, then

$$A =_c g f[A] \subseteq g[C] \subseteq A, \quad g[C] =_c C.$$